



Cesar Yaril Velazquez Vazquez

SENIOR SOFTWARE ENGINEER | APPLIED AI & FULL-STACK DEVELOPMENT

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SKILLS

Applied AI / LLM Applications

AI Agents & Agentic Workflows

.NET / C#

RAG & Tool Calling

AWS / Docker / CI/CD

LANGUAGES

English

Español

PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience building enterprise, cloud, and web applications using .NET/C#, Java, Angular, React, AWS, and Docker. Experienced in integrating AI tools into production development workflows, including AI agents, agentic workflows, automated testing, and LLM-powered applications. Focused on Applied AI systems that combine LLMs with enterprise data, APIs, deterministic business logic, evaluation, and guardrails.

EMPLOYMENT HISTORY

Full stack developer at Alset

Jan 2025 — Present

Develop and maintain an AWS-hosted enterprise application using .NET, frontend technologies, Docker, and automated validation workflows. Integrated AI development tools, AI agents, and agentic workflows to support software analysis, testing, debugging, and development tasks across the application lifecycle.

Full-Stack Developer at Volante Technologies

Jan 2021 — Dec 2024

- Developed enterprise software for financial institutions, including financial dashboards, cloud-based applications, and AI-assisted tools. Enhanced chatbot capabilities and used AI development tools while collaborating with international engineering teams.

Lead Developer at Tesla Software

Jan 2017 — Dec 2021

- Led the development of large-scale, cross-platform applications.
- Incorporated AI-driven services to enhance functionality.
- Delivered clean, user-focused interfaces with responsive design.
- Managed continuous integration and deployment pipelines.

Lead Developer at IIASA

Jan 2014 — Dec 2017

- Directed multiple web-based enterprise projects.
- Developed fast-loading dashboards and data visualization tools.
- Mentored junior developers and managed version control.

Web Developer at DA7OS

- Created responsive websites and backend systems for small businesses.
- Built inventory, CRM, and HR modules with a focus on usability.

SELECTED APPLIED AI PROJECTS

Autonomous AI Agent for E2E Test Generation & Self-Repair

Designed and implemented an AI testing agent that analyzes a legacy ASP.NET MVC 5 enterprise application, autonomously generates Playwright/TypeScript E2E tests, executes them, diagnoses technical failures, and performs controlled test repairs.

- Generates smoke, navigation, validation, and complete CRUD test flows from application context.
- Analyzes controllers, views, dynamic forms, and Syncfusion components to construct relevant test scenarios.
- Automatically repairs selectors, AJAX synchronization issues, test-data formats, and other recoverable test failures.
- Uses structured outputs, reusable Page Objects, temporary test data, and automated cleanup.
- Provides execution tracing, diagnostics, agent decisions, and automated Playwright HTML reports.
- Separates safe and destructive test operations, allowing CRUD actions only in authorized local environments.

Technologies: TypeScript, Playwright, Node.js, Codex CLI, LLM Agents, Structured Outputs, Guardrails, Evals, Page Object Model, ASP.NET MVC 5, Syncfusion.

Applied AI Engineer — Multi-Agent Educational Analytics System

Designed and implemented an agentic AI architecture that transforms academic and financial data into explainable, evidence-backed insights by combining deterministic analytics, LLM reasoning, tool calling, contextual retrieval, and stateful workflows.

- Orchestrates specialized agents for weekly summaries, student 360° profiles, group health, campus indicators, and financial analysis.
- Uses LangGraph for state management, routing, tool execution, retries, and human-in-the-loop workflows.
- Exposes typed tools over existing REST APIs and SQL repositories with user, role, and campus-level authorization.
- Combines deterministic SQL/C# analytics with LLM interpretation of trends, anomalies, and correlations.
- Uses RAG to retrieve institutional policies, manuals, and internal documentation when additional context is required.
- Produces structured, schema-validated outputs linked directly to supporting evidence.
- Implements guardrails against unsupported claims, unauthorized actions, and sensitive-data exposure.
- Includes contextual memory, tracing, quality evaluations, and cost/latency monitoring.
- Supports conversational analytics, proactive alerts, and automated management reporting.

Technologies: Python, LangGraph, LangChain, OpenAI API, Tool Calling, RAG, Embeddings, Vector Databases, C#, ASP.NET Core, SQL Server, REST APIs, Pydantic, Agent Observability & Evals.

EDUCATION

Degree in Computer Science at Universidad Autónoma del Estado de Morelos